

L9C and T17C in the CHGA leader of engineered insulin

Computational analysis of Kobaisi et al. (2026) | 15 September 2026

Candidate: introduce L9C + T17C into the CHGA-derived leader of the engineered INS construct. The two substitutions lower predicted HLA-I presentation counts while preserving the predicted cleavage site and the proinsulin sequence.

Sequence

Baseline: MRSAAVLALLLCAGQVTA + INS25-110

Candidate: MRSAAVLACLLCAGQVCA + INS25-110

Coordinates refer to the CHGA leader. Calculations assume CHGA1-18 + INS25-110, a 104-residue precursor reconstructed from canonical sequences. The exact synthesized construct in Kobaisi et al. was unavailable; sequence agreement requires confirmation.

Quantitative results

Counts below each strict percentile cutoff, baseline → candidate:

Predictor and context	<0.5%	<1%	<2%	Reduction at <2%
NetMHCpan EL	4 → 0	8 → 2	17 → 8	52.9%
MHCflurry with flanks	9 → 3	19 → 6	27 → 14	48.1%
MHCflurry without flanks	4 → 1	12 → 5	21 → 12	42.9%

Denominator: 504 leader/junction peptide-window/HLA pairs, comprising 72 overlapping 8-11mer windows starting at precursor positions 1-18 × seven HLA alleles. These are predicted-presentation counts, not measured immune-escape percentages.

Full precursor: at <2%, NetMHCpan counts decrease 59 → 50 (**15.3%**) and MHCflurry with flanks 72 → 59 (**18.1%**), across 2,674 pairs. Downstream INS peptide sequences remain unchanged.

Selection: 152 single substitutions and eight selected double variants were screened. L9C + T17C was the only tested double with no count increase in any of 42 comparisons (seven alleles × two models × three cutoffs) under both flank settings: 16 decreased / 26 unchanged with flanks; 15 / 27 without.

Cleavage: SignalP 6.0 predicts the same 18/19 boundary and INS25-110 product. Cleavage probabilities are 0.979955 → 0.979581. The added cysteines have not been assessed for folding, secretion or insulin activity.

Residual predictions

Three HLA-C*03:04 pairs remain below the MHCflurry 0.5% cutoff: RSAAVLACL, SAAVLACLL and CAFVNQHL. The A*02:01 peptide VLACLLCAGQV has stronger predicted NetMHCpan binding than its baseline counterpart, 654.57 → 202.39 nM, while EL rank remains 5.7%.

Suggested comparison: parental versus L9C + T17C constructs, measuring HLA-I ligand presentation alongside proinsulin processing, secretion and insulin activity.

Methods and accompanying files

HLA panel: A*02:01, A*03:01, A*24:02, B*40:01, B*49:01, C*03:04, C*07:01. Predictors: NetMHCpan 4.1, MHCflurry 2.2.1 with 2.2.0 models, SignalP 6.0. Flank comparisons use available five-residue flanks or none. Accompanying data contain 34,762 prediction rows for 13 constructs, sequences and cached outputs; python3 reproducibility/reproduce.py reproduces the reported counts offline.

Source study: Kobaisi et al., Cell Reports 45, 117941 (2026).

Computational analysis and drafting used Codex agents; predictor outputs and analysis scripts are supplied for review.